

Abhimanyu Saklani

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CAREER SUMMARY

Computer Science undergraduate with hands-on experience in AI, Natural Language processing, Machine Learning, and Data Science, proficient in Python, SQL, and Scikit-learn. Experienced in building end-to-end ML pipelines, predictive models, statistical analysis, feature engineering, and data-driven applications. Passionate about solving real-world problems through scalable and data driven solutions.

EDUCATION

Nitte Meenakshi Institute of Technology (NMIT) B.E. in Computer Science & Engineering CGPA: 9.24 / 10.0	<i>Bengaluru, KA</i> 2023 – 2027
Jawahar Navodaya Vidyalaya, Mandi, HP Higher Secondary (CBSE) Percentage: 90.8%	<i>Mandi, HP</i> 2023

EXPERIENCE

Commonwealth Bank of Australia Software Developer Trainee (Apprenticeship)	<i>Remote</i> July, 2026
- Completing a structured software development apprenticeship under the National Apprenticeship Promotion Scheme. - Gaining hands-on exposure to enterprise software development practices and workflows.	

TECHNICAL SKILLS

Languages: Python, Java, SQL
Relevant Coursework: Artificial Intelligence and Machine Learning, Deep Learning, Data Science using Python, Data Mining, Data Structures, Big Data Technologies, Database Management System Concepts, Object Oriented Programming (OOPs).
AI & Generative AI: GPTs, LLMs, Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Agents, LangChain.
ML & Data Science: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Linear Regression, Logistic Regression, XGBoost, PyTorch, Tensorflow, Feature Engineering, Statistical Analysis, Correlation analysis, NLP, Hypothesis Testing, Classification, Clustering, PCA, Tableau, Power BI.
Backend and Cloud: FastAPI, REST APIs, PostgreSQL, MongoDB, MySQL, Microsoft Azure.
Tools: GitHub, GitHub Actions, Git, Jupyter Notebook, Google Colab, Google Workspace, VS Code

PROJECTS

- AI-Based Spark Job Performance Optimization System** | *Python, PySpark, XGBoost, Pandas*
- Cut Spark job execution time by **64–72%**, measured across three enterprise-scale workload profiles, by developing an XGBoost ML pipeline that learns from PySpark execution parameters and outputs optimised job configurations.
 - Achieved prediction accuracy ($R^2 = 0.976$) on job runtime forecasting, validated on a held-out production-mirroring test set, by engineering domain-specific features from Spark execution logs and tuning model hyperparameters.
 - Reduced infrastructure cost exposure for data engineering teams by integrating the recommendation engine as a configurable module callable from existing data pipeline orchestration tools, cutting manual configuration effort per job run.
- Cardiovascular Disease Prediction** | *Python, Scikit-learn, XGBoost, CatBoost, SHAP, Pandas*
- Built an end-to-end predictive analytics system for cardiovascular disease detection using supervised machine learning on 68000 cleaned patient records.
 - Engineered clinically relevant features, implemented data preprocessing, feature selection, model evaluation, and 10-fold cross-validation to ensure robust model generalization.
 - Optimized XGBoost and CatBoost classifiers through threshold tuning, achieving **0.803 ROC-AUC**, **83.0% Recall**, and **74.2% F1-score**, while leveraging **SHAP** for interpretable AI-driven predictions.
- Document Intelligence System** | *Python, LangChain, Groq API, VectorDB*
- Increased document QA response relevance by **18%**, measured via RAGAS evaluation metrics, by building a RAG pipeline with multi-stage chunking, embedding generation, and semantic retrieval using LangChain and Groq API.
 - Enabled question answering over **400+ page** unstructured document corpora, tested on real regulatory and technical documents, by designing ingestion, preprocessing, and retrieval pipelines supporting both structured and unstructured data sources.
 - Improved retrieval precision through reranking, validated with automated scoring, by implementing retrieval optimisation loops.

CERTIFICATIONS

Google Advanced Data Analytics Professional Certificate Issued by Google	2025
- Validated skills in data wrangling, statistical analysis, Python-based ML pipelines, and communicating data-driven insights.	
Citizen Data Science using Python Issued by Infosys Springboard	2025
- Validated EDA, data visualization, regression analysis, time series analysis, association analysis, and ML fundamentals.	
Python for Data Science - IIT Madras Issued by NPTEL	2024
- Validated Python-based data science workflows covering NumPy, Pandas, Matplotlib, and introductory ML techniques.	

ACHIEVEMENTS

1st Prize, Sustainability Track, Vibe-a-thon- contributed AI integration for "[Eco-Tracker](#)", an AI-powered smart-city sustainability platform; built the AI-driven "Smart Sort" feature that classifies waste from photos to guide recycling.
First Runner Up, CRYPTEC Coding Battle 2025, Department of Computer Science, NMIT 2025